Braid Moves in Commutation Classes of the Symmetric Group

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Outline

1. Introduction
2. Reduced Words and Standard Shifted Staircase Tableaux
3. Right-Justified Tableaux
4. Half-Right-Justified Tableaux
5. Homomesy

Outline

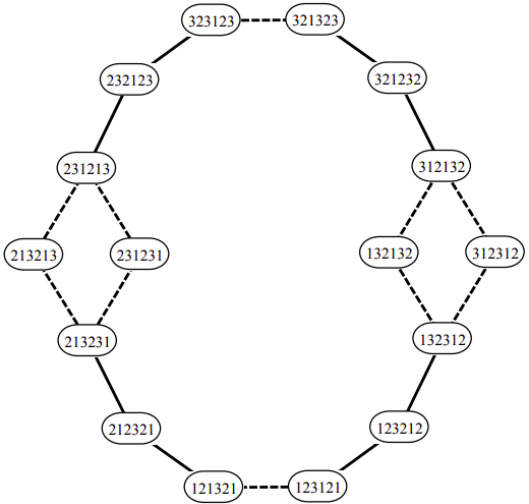
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Definition

- $\mathfrak{S}_n :=$ symmetric group of degree n
- $s_i := (i, i+1)$, transposition that swaps $i, i+1$
- $S := \{s_i : 1 \leq i \leq n-1\}$, a generating set of $\mathfrak{S}_n$
- s_i's properties:
  - $s_i^2 = 1$
  - $s_i s_j = s_j s_i$ for $|i-j| > 1$
  - $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$, this is called the **braid move**
- Let $w \in \mathfrak{S}_n$, $\ell(w) :=$ the smallest ℓ s.t. w can be composed by $s_{i_1} s_{i_2} \cdots s_{i_\ell}$.
- Suppose $w = s_{i_1} s_{i_2} \cdots s_{i_\ell}$, $\mathbf{w} = i_1 i_2 \cdots i_\ell$ is called a **word** for w. If $\ell = \ell(w)$, then $\mathbf{w}$ is called a **reduced word**.
- $\mathrm{Red}(w) :=$ the graph formed by the reduced words of w with edges given by commutations ($s_i s_j = s_j s_i$ for $|i-j| > 1$) and braid moves.
- $w_0 :=$ the element in $\mathfrak{S}_n$ with the largest $\ell(w_0)$
- $w_0 = (s_1 s_2 \cdots s_{n-1})(s_1 s_2 \cdots s_{n-2}) \cdots (s_1 s_2)(s_1)$

Example of $\mathrm{Red}(w)$

- $\mathrm{Red}(w_0)$ for $\mathfrak{S}_4$: (solid lines: braid moves; dotted lines: commutations)



V. Reiner Theorem

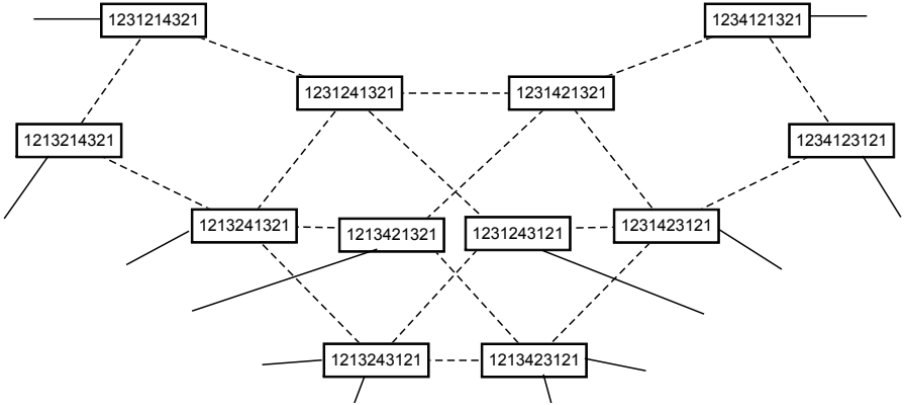
- Question: How many braid moves are there in $\mathrm{Red}(w)$?
- V. Reiner: The expected number of braid moves for a reduced word for $w_0 \in \mathfrak{S}_n$ is 1. (i.e. $\frac{1}{2}|\mathrm{Red}(w_0)|$ edges in $\mathrm{Red}(w_0)$ are braid moves.)

Commutation Classes

- $\mathbf{w}$: a reduced word of $w \in \mathfrak{S}_n$
- $\mathrm{Red}(\mathbf{w})$: induced subgraph of $\mathrm{Red}(w)$ with reduced words connected to $\mathbf{w}$ using only commutations
- $\mathbf{w}_0 := (s_1 s_2 \cdots s_{n-1})(s_1 s_2 \cdots s_{n-2}) \cdots (s_1 s_2)(s_1)$, a reduced word of w_0

Example Commutation Class

- $\mathrm{Red}(\mathbf{w}_0)$ for $\mathfrak{S}_5$: (solid lines: braid moves; dotted lines: commutations)



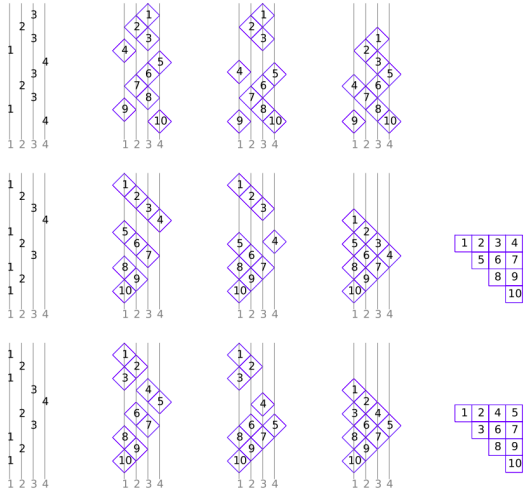
Theorem 1.2

- Question: How many braid moves are there in $\mathrm{Red}(\mathbf{w})$?
- Theorem: The expected number of braid moves for a reduced word in the commutation class of the word $\mathbf{w}_0$ is 1.

Outline

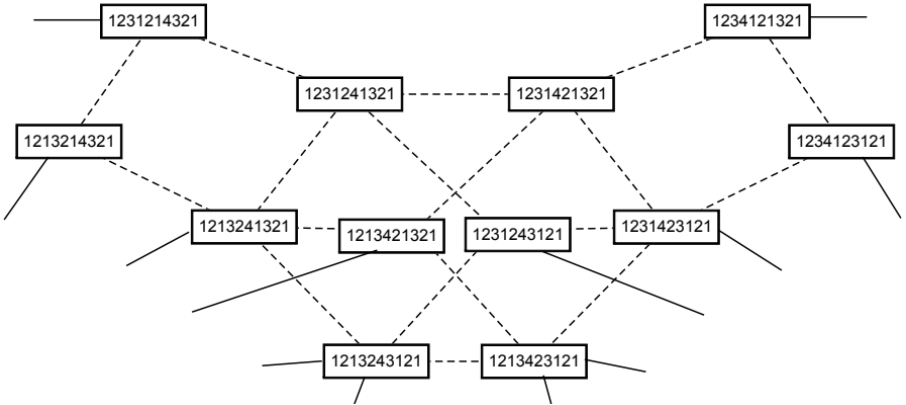
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Bijection between Reduced Words and Standard Shifted Staircase Tableaux



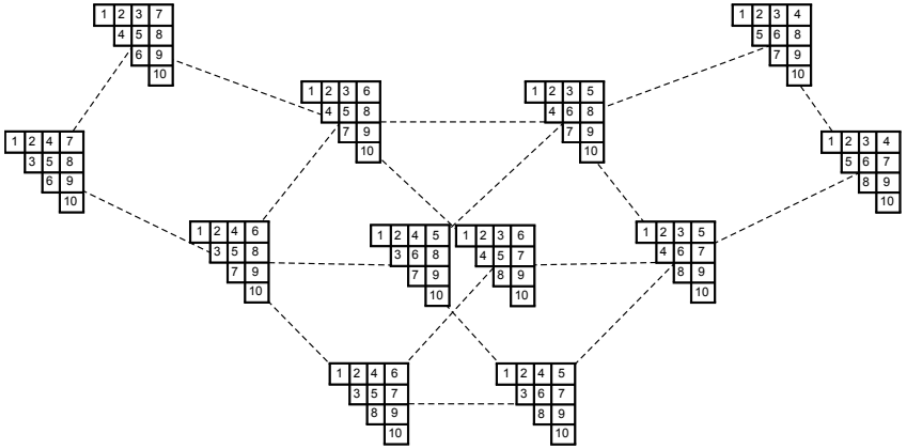
Bijection between Reduced Words and Standard Shifted Staircase Tableaux

- $\mathrm{Red}(\mathbf{w}_0)$:



Bijection between Reduced Words and Standard Shifted Staircase Tableaux

- $\mathrm{Red}(\mathbf{w}_0)$:

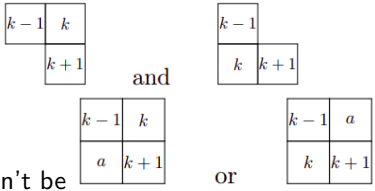


Bijection between Reduced Words and Standard Shifted Staircase Tableaux

- $\mathrm{ShSYT}(\Delta_n) :=$ the set of all standard shifted staircase tableaux of shape $\Delta_n (:= (n, n-1, \ldots, 1))$
- Why is it a standard shifted tableau?
- Why is it always staircase?
- Lemma: $\mathbf{w} \in \mathrm{Red}(\mathbf{w}_0)$ if and only if:
  - $\mathbf{w}$ contains exactly i s_i, and
  - Between every two s_i, there is an s_{i+1}.

Lemma 2.2

- The only possible braid moves in elements of $\mathrm{Red}(\mathbf{w}_0)$ are those of the form $s_1 s_2 s_1$.
- Proof:
  - Braid moves $s_i s_{i+1} s_i$ and $s_{i+1} s_i s_{i+1}$:



  - It shouldn't be because $k-1 < a < k+1$ but $a \neq k$.

  - The only possible hook is $s_1 s_2 s_1$.
  - Call this kind of hooks **braid hooks**.
- Theorem 1.3: The expected number of braid hooks of elements in $\mathrm{ShSYT}(\Delta_n)$ is 1.
- By lemma 2.2, theorem 1.3 is equivalent to theorem 1.2.

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Definition

- $\mathrm{rSYT}(\lambda) :=$ the set of standard tableaux of the diagram given by λ, where all rows are right-justified
- $\lambda = (5, 2, 1)$: (where the blue numbers are the braid hooks)

$$\mathsf{rSYT}(\lambda) = \left\{ \begin{array}{|c|c|c|c|c|}\hline 1 & 2 & 3 & 4 & 5 \\ \hline & & & 6 & 7 \\ \cline{4-5} & & & & 8 \\ \cline{5-5} \end{array} , \begin{array}{|c|c|c|c|c|}\hline 1 & 2 & 3 & 4 & 6 \\ \hline & & & 5 & 7 \\ \cline{4-5} & & & & 8 \\ \cline{5-5} \end{array} \right\}$$

Expected number of braid hooks $= 1$?

Theorem 3.2

- Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_\ell)$ be a partition such that $\lambda_1 > \lambda_2$ and $\lambda_\ell = 1$. Then the expected number of braid hooks in $\mathrm{rSYT}(\lambda)$ is 1.
- Proof: Construct a bijection $\varphi : \{(k, t) : k \text{ a braid hook in } t \in \mathrm{rSYT}(\lambda)\} \to \mathrm{rSYT}(\lambda)$.

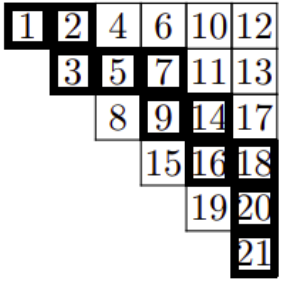
Partial (Inverse) Promotion Operators

- $\tau_i : \mathrm{rSYT}(\lambda) \to \mathrm{rSYT}(\lambda)$
  $t \to t.\tau_i$
- $\tau_i :=$ interchanges i and $i+1$ in t if the result is again in $\mathrm{rSYT}(\lambda)$; otherwise leaves t fixed.
- $\partial_i := \tau_i \tau_{i+1} \cdots \tau_{|\lambda|-1}$, $\partial_i^* := \tau_{i-1} \tau_{i-2} \cdots \tau_1$
- $\partial := \partial_1$: **promotion operator**, $\partial^* := \partial_{|\lambda|}^*$: **inverse promotion operator**, ∂_i: **partial promotion operator**, $\partial_{|\lambda|}^*$: **partial inverse promotion operator**
- $\varphi(k, t) := t.\partial_k^* \partial_k$
- What do promotion operators do?

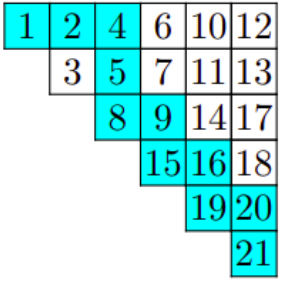
(Inverse) Promotion Sliding Path

- $\mathcal{L}$: promotion sliding path, $\mathcal{R}$: inverse promotion sliding path

$\mathcal{L}$:



$\mathcal{R}$:



Construction of φ^{-1}

- Given $t \in \mathrm{rSYT}(\lambda)$, find unique $t' \in \mathrm{rSYT}(\lambda)$ and k s.t. $\varphi(t', k) = t$.
- τ_i is a bijection, and so are ∂_k, ∂_k^*
- $t' = t.(\partial_k^* \partial_k)^{-1}$, and is determined by $k \Rightarrow$ we only need to prove the uniqueness of k
- k is a braid hook of t', so it should be 

.

Uniqueness of k

- There are no 

or 

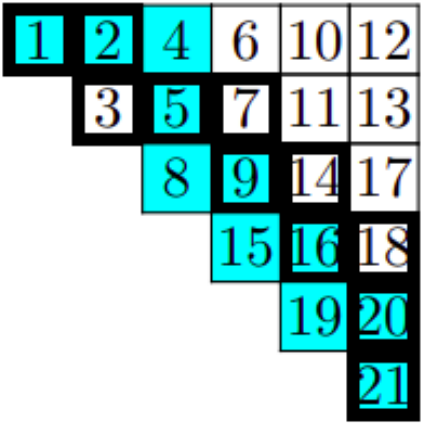
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- 

: $\mathcal{R}$ is (locally) above $\mathcal{L}$.
- 

: $\mathcal{L}$ is (locally) above $\mathcal{R}$.
- The northeastern path is (locally) above the other.
- Start: $\mathcal{R}$ being above $\mathcal{L}$; finish: $\mathcal{L}$ being above $\mathcal{R}$; such crossing happens for exactly one time.

Uniqueness of k

$t =$



Uniqueness of k

- $\Rightarrow k = 5$ and

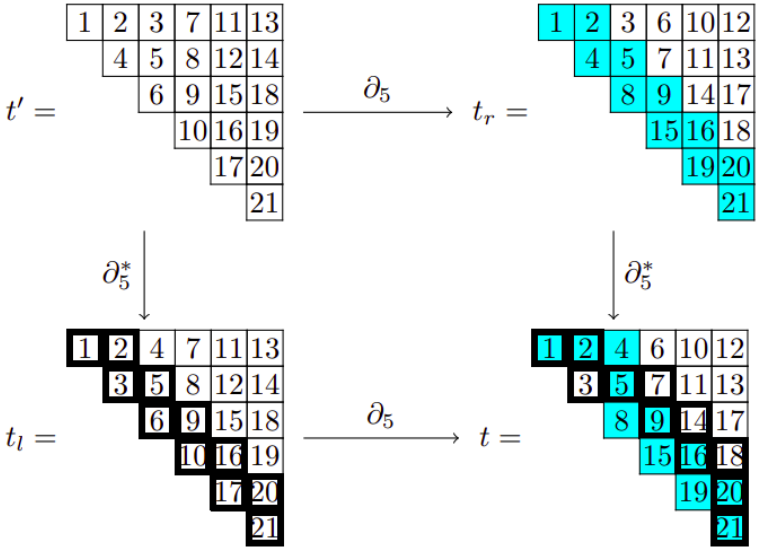
$t' =$

1	2	3	7	11	13
	4	5	8	12	14
		6	9	15	18
			10	16	19
				17	20
					21

Theorem 3.2

- We've found φ^{-1}, and theorem 3.2 is proved.

Corollary to Theorem 3.2



Corollary to Theorem 3.2

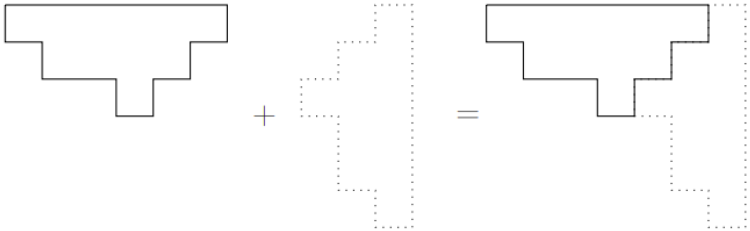
- **Left partial braid hook**: a hook with values $(*, k, k+1)$ which lies on the promotion path
- **Right partial braid hook**: a hook with values $(k-1, k, *)$ which lies on the inverse promotion path
- There are bijections between:
  - (k, t'), where $t' \in \mathrm{rSYT}(\lambda)$ and k is a braid hook of t'
  - (k, t_l), where $t_l \in \mathrm{rSYT}(\lambda)$ and k is a left partial braid hook of t_l
  - (k, t_r), where $t_r \in \mathrm{rSYT}(\lambda)$ and k is a right partial braid hook of t_r
  - t, where $t \in \mathrm{rSYT}(\lambda)$
- The expected number of left partial braid hooks in all tableaux in $\mathrm{rSYT}(\lambda)$ is 1.

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Definition

- $\lambda : \lambda_1 > \lambda_2 > \cdots > \lambda_\ell$
- The rightmost cell of each row is one step below and to the left of the rightmost cell of the previous row.
- $\mathrm{hrSYT}(\lambda) :=$ the set of half-right-justified SYT of shape λ



- $\Delta_n^t := (2n+1, 2n-1, \ldots, 5, 3, 1)$.

Theorem 4.2

- The expected number of braid hooks in $\mathrm{hrSYT}(\lambda)$ is at most $\frac{1}{2}$. If $\lambda_1 \geq \lambda_2 + 2$ and $\lambda_\ell = 1$, then the expected number of braid hooks in $\mathrm{hrSYT}(\lambda)$ is exactly $\frac{1}{2}$.

Directly Import Proof of Theorem 3.2

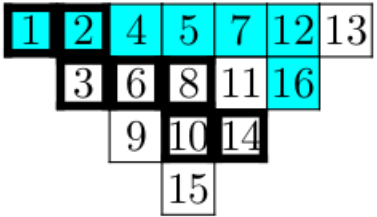
- Proof:
  - Construct $\psi : \{(k, t) : k \text{ a braid hook in } t \in \mathrm{hrSYT}(\lambda)\} \to \mathrm{hrSYT}(\lambda)$.
  - $\psi(k, t) := t.\partial_k^* \partial_k$.
  - Similarly define the promotion and the inverse promotion sliding path $\mathcal{L}, \mathcal{R}$.
  - k is a braid hook of $t.(\partial_k^* \partial_k)^{-1}$, it should be 

.
  - There are no 

or 

.
  - $\mathcal{L}$ and $\mathcal{R}$ must cross for exactly once. Therefore, ψ is a bijection and the expected number of braid hooks $= 1$.
  - What goes wrong in this proof?

Fix the Proof

- Start: $\mathcal{R}$ being above $\mathcal{L}$; but finish: $\mathcal{L}$ not always being above $\mathcal{R}$.
- $\mathcal{L}$ and $\mathcal{R}$ will cross for at most one time.
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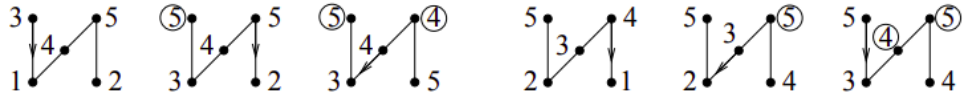
- ψ is injective but not bijective.
- The sufficient conditions of theorem 4.2:
  - $\mathcal{L}$ and $\mathcal{R}$ cross in at most half of the tableaux in $\mathrm{hrSYT}(\lambda)$.
  - $\mathcal{L}$ and $\mathcal{R}$ cross in exactly half of the tableaux in $\mathrm{hrSYT}(\lambda)$ if $\lambda_1 \geq \lambda_2 + 2$ and $\lambda_\ell = 1$.

(Dual) Evacuation Maps

- $\tau_i : \mathrm{hrSYT}(\lambda) \to \mathrm{hrSYT}(\lambda)$
- $t \to t.\tau_i$
- $\tau_i :=$ interchanges i and $i+1$ in t if the result is again in $\mathrm{hrSYT}(\lambda)$; otherwise leaves t fixed.
- $\epsilon := (\tau_1\tau_2\cdots\tau_{|\lambda|-1})(\tau_1\tau_2\cdots\tau_{|\lambda|-2})\cdots(\tau_1\tau_2)(\tau_1)$ is called the **evacuation** map.
- $\epsilon^* := (\tau_{|\lambda|-1}\tau_{|\lambda|-2}\cdots\tau_1)(\tau_{|\lambda|-1}\tau_{|\lambda|-2}\cdots\tau_2)\cdots(\tau_{|\lambda|-1}\tau_{|\lambda|-2})(\tau_{|\lambda|-1})$ is called the **dual evacuation** map.

(Dual) Evacuation Maps

- $\epsilon^2 = \epsilon^{*2} = 1$?



and

- Theorem 4.2 is equivalent to:
  - $\mathcal{L}$ and $\mathcal{R}$ cross in at most one of $t, t.\epsilon$ for all $t \in \mathrm{hrSYT}(\lambda)$.
  - $\mathcal{L}$ and $\mathcal{R}$ cross in exactly one of $t, t.\epsilon$ for all $t \in \mathrm{hrSYT}(\lambda)$ if $\lambda_1 \geq \lambda_2 + 2$ and $\lambda_\ell = 1$.

Conjugate

- $t^\dagger$ is called the **conjugate** of t.

$$t = \begin{array}{|c|c|c|c|c|}\hline 1 & 2 & 3 & 4 & 9 \\ \hline \end{array}\!\!\begin{array}{|c|c|c|c|}\hline 5 & 6 & 8 \\ \hline \end{array}\!\!\begin{array}{|c|}\hline 7 \\ \hline \end{array} \longrightarrow \begin{array}{|c|}\hline 9 \\ \hline \end{array}\!\!\begin{array}{|c|c|}\hline 8 & 4 \\ \hline \end{array}\!\!\begin{array}{|c|c|c|}\hline 7 & 6 & 3 \\ \hline \end{array}\!\!\begin{array}{|c|c|}\hline 5 & 2 \\ \hline \end{array}\!\!\begin{array}{|c|}\hline 1 \\ \hline \end{array} \longrightarrow \begin{array}{|c|}\hline 1 \\ \hline \end{array}\!\!\begin{array}{|c|c|}\hline 2 & 6 \\ \hline \end{array}\!\!\begin{array}{|c|c|c|}\hline 3 & 4 & 7 \\ \hline \end{array}\!\!\begin{array}{|c|c|}\hline 5 & 8 \\ \hline \end{array}\!\!\begin{array}{|c|}\hline 9 \\ \hline \end{array} = t^\dagger$$

Relations between the Operators

By definition it is clear that

- $\partial^* = \partial^{-1}$
- $\dagger^2 = 1$

Notice that $\dagger\tau_i\dagger = \tau_{|\lambda|-i}$

- $\dagger\partial\dagger = \partial^*$
- $\dagger\epsilon\dagger = \epsilon^*$

We can convert $\epsilon = (\tau_1\tau_2\cdots\tau_{|\lambda-1|})(\tau_1\tau_2\cdots\tau_{|\lambda-2|})\cdots(\tau_1\tau_2)\tau_1$ to $\tau_1(\tau_2\tau_1)\cdots(\tau_{|\lambda-2|}\cdots\tau_2\tau_1)(\tau_{|\lambda-1|}\cdots\tau_2\tau_1)$ only by swapping neighboring τ_i, τ_j with $|i-j| > 1$. And this leads to

- $\epsilon^2 = (\epsilon^*)^2 = 1$
- $\epsilon\partial = \partial^*\epsilon$
- $\epsilon^*\partial = \partial^*\epsilon^*$

Staircase Pair

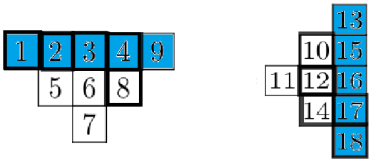
- Given $t \in \mathrm{hrSYT}(\lambda)$, define the *staircase pair* $(t, (t.\epsilon)^\dagger) \in \mathrm{SYT}$ as follows.

$$t = \begin{array}{|c|c|c|c|c|}\hline 1&2&3&4&9\\\hline\cline{2-4} 5&6&8\\\cline{2-4}\cline{3-3} 7\\\cline{3-3}\end{array} \qquad t.\epsilon = \begin{array}{|c|c|c|c|c|}\hline 1&2&3&4&6\\\hline\cline{2-4} 5&7&9\\\cline{2-4}\cline{3-3} 8\\\cline{3-3}\end{array}, \qquad (t.\epsilon)^\dagger = \begin{array}{|c|c|c|}\hline &&4\\\hline\cline{2-3} &1&6\\\cline{2-3} 2&3&7\\\cline{2-3} 5&8\\\cline{2-2} 9\\\cline{2-2}\end{array}, \qquad (t.\epsilon)^\dagger + 9 = \begin{array}{|c|c|c|}\hline &&13\\\hline\cline{2-3} &10&15\\\cline{2-3} 11&12&16\\\cline{2-3} 14&17\\\cline{2-2} 18\\\cline{2-2}\end{array},$$

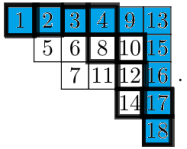
$$(t, (t.\epsilon)^\dagger) = \begin{array}{|c|c|c|c|c|c|}\hline 1&2&3&4&9&13\\\hline\cline{2-5} 5&6&8&10&15\\\cline{2-5}\cline{3-4} 7&11&12&16\\\cline{3-5} 14&17\\\cline{4-5} 18\\\cline{4-4}\end{array}.$$

Promotion Paths of Staircase

- Observe the promotion and inverse promotion paths $\mathcal{L}_1, \mathcal{R}_1$ of t, and $\mathcal{L}_2, \mathcal{R}_2$ of $(t.\epsilon)^\dagger$



- Then observe the promotion and inverse promotion paths $\mathcal{L}_s, \mathcal{R}_s$ of $(t, t.\epsilon)^\dagger)$



- As we can see, $\mathcal{L}_s = \mathcal{L}_1 \cup \mathcal{L}_2$ and $\mathcal{R}_s = \mathcal{R}_1 \cup \mathcal{R}_2$ in this case, is this a coincidence?

Lemma 4.14

- Lemma 4.14 : The restriction of $\mathcal{L}_s$ on $t, (t.\epsilon)^\dagger$ are $\mathcal{L}_1, \mathcal{L}_2$ respectively. Similarly, the restriction of $\mathcal{R}_s$ on $t, (t.\epsilon)^\dagger$ are $\mathcal{R}_1, \mathcal{R}_2$ respectively.
- It is clear that the restriction of $\mathcal{L}_s$ on t is $\mathcal{L}_1$ and that the restriction of $\mathcal{R}_s$ on $(t.\epsilon)^\dagger$ is $\mathcal{R}_2$
- To prove that the restriction of $\mathcal{R}_s$ on t is R_1, it remains to prove that $\mathcal{R}_s$ passes the cell containing $|\lambda|$.

Proof of Lemma

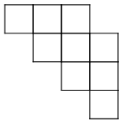
- Lemma 4.10 : If $|\lambda|$ is contained in the cell (i, j) in t, then the promotion path $\mathcal{L}$ of $(t.\epsilon)$ also ends on the cell(i, j).
- Proof: Notice that the end of the promotion path is the cell containing $|\lambda|$ in $(t.\epsilon).\partial$
- We have previously stated that $\epsilon\partial = \partial^*\epsilon = (\tau_1\tau_2 \dots \tau_{|\lambda|-2}) \cdots (\tau_1\tau_2)\tau_1$, which does not affect the position of $|\lambda|$. So $|\lambda|$ is contained in the cell in $t.\epsilon\partial$ and in t, which proves lemma 4.10.
- The promotion path of $(t.\epsilon)$ correspond to $\mathcal{R}_2$ in $(t.\epsilon)^\dagger$, lemma 4.10 proves that $\mathcal{R}_2$ ends at the cell next to $|\lambda|$, $\mathcal{R}$ must pass $|\lambda|$ because it is the greatest number in t. Lemma 4.14 is thus proved.

Proof of theorem

- $(t, (t.\epsilon)^\dagger)$ is a SYT, so $\mathcal{L}_s$ and $\mathcal{R}_s$ cross at most once, and lemma 4.14 tells us that $\mathcal{L}_1$ and $\mathcal{R}_1$ crosses if and only if $\mathcal{L}_s$ and $\mathcal{R}_s$ cross inside t.
  Likewise, $\mathcal{L}_2$ and $\mathcal{R}_2$ crosses if and only if $\mathcal{L}_s$ and $\mathcal{R}_s$ cross inside $(t.\epsilon)^\dagger$, clearly at most one of the events could happen.
- Proposition : If the paths $\mathcal{L}$ and $\mathcal{R}$ of t cross, then in $(t.\epsilon)$, the paths do not cross. And the converse also holds if $\lambda_1 \geq \lambda_2 + 2$ and $\lambda_\ell = 1$, that is, $\mathcal{L}_s$ and $\mathcal{R}_s$ must cross inside exactly one of the subtableaux.
- Proof : The first condition correspond to the condition $\lambda_1 > \lambda_2$ in the case of SYT, which guarantees the existence of the cross, and the second condition prevent that the braid hook occurs at the intersection of t and $(t.\epsilon)^\dagger$, so they must cross exactly once inside one of the subtableaux.

Skew Right-Justified Shape

- Let $\mu \subset \lambda$ and consider the standard tableaux of skew right-justified shape λ/μ, denoted by $\mathrm{rSYT}(\lambda/\mu)$.
- Suppose $\lambda_1 > \lambda_2$ and λ_ℓ still holds and λ/μ is connected, then if we draw again the paths $\mathcal{L}, \mathcal{R}$, then by similar deduction, they must cross somewhere, but may cross more than once.
- Here's an example of $\lambda/\mu = (4,3,2,1)/(1)$, the paths $\mathcal{L}$ and $\mathcal{R}$ cross three times.





Proposition 4.17

- Like the case in $\mathrm{rSYT}(\lambda)$, $\mathcal{L}$ is above $\mathcal{R}$ at first and $\mathcal{R}$ is above $\mathcal{L}$ in the end.
- Proposition 4.17: If λ and μ satisfy the previous conditions, then $\mathcal{L}, \mathcal{R}$ must cross at least once, and the difference between the number of crossings ($\mathcal{R}$ above $\mathcal{L}$) to ($\mathcal{L}$ above $\mathcal{R}$) minus the number of the opposite crossings, is one.
- Translate back to the words, for a communication class corresponding to a skew shape, the expected difference between the number of braid moves of the form $s_i s_{i+1} s_i$ and the number of braid moves of the form $s_{i+1} s_i s_{i+1}$ is one.

Outline

1. Introduction
2. Reduced Words and Standard Shifted Staircase Tableaux
3. Right-Justified Tableaux
4. Half-Right-Justified Tableaux
5. Homomesy

Definition

- We use another approach to prove the theorem 3.2 and showing that the number of braid hooks has a **homomesy** property.
- Definition : For a set S and a group action τ on S, we say a function $f : S \to K$ is **homomesic** if for any τ-orbit $\mathcal{O} \subset S$, $\frac{1}{\#\mathcal{O}} \sum_{x \in \mathcal{O}} f(x)$ is a constant in K, where K is a field.
- To demonstrate the homomesy, we need to first define a group action.

Odd and Even promotion operators

- Let $\tau_o = \prod_{i \text{ odd}} \tau_i$ and $\tau_e = \prod_{i \text{ even}} \tau_i$ be the *odd* and even promotion operator. And let $G = \langle \tau_o, \tau_e \rangle$
- Notice that both τ_o and τ_e are involutions, and that $G = \langle \tau_o, \tau_o\tau_e \rangle$ where $1 = \tau_o^2 = (\tau_o(\tau_o\tau_e))^2$, therefore G is a dihedral group.
- Theorem 5.1. The number of braid hooks is homomesic with respect to G on $\mathrm{rSYT}(\lambda)$ if and only if $\lambda_1 > \lambda_2$ and $\lambda_\ell = 1$.

Theorem 5.2

We prove the result in terms of reduced words instead.

- Define $\mathrm{rW}(\lambda)$ =the communication class of reduced words corresponding to $\mathrm{rSYT}(\lambda)$ in Viennot's bijection.
- Theorem 5.2. The expected value of the number of braid moves in $\mathrm{rW}(\lambda)$ is at most one, and it is exactly one if and only if $\lambda_1 > \lambda_2$ and $\lambda_\ell = 1$.
  And in this case, the braid moves number is homomesic to the group $\langle \tau_o, \tau_e \rangle$, which acts correspondingly on $\mathrm{rW}(\lambda)$

Construct the map Φ

- To prove this property, the previous bijection used in previous proof is harder to use because the map does not preserve orbits.
- Define another map Φ as follows

$$\Phi : \{(k, \mathbf{w}) | \mathbf{w} \in \mathrm{rW}(\lambda), k \text{ is a braid in } \mathbf{w}\} \to \mathrm{rW}$$

$$\Phi(k, \mathbf{w}) = \mathbf{w}.\tau_{o(k-2)} \cdots \tau_{o(1)}$$

  where $\tau_{o(i)} = \tau_o$ if i is odd and otherwise is τ_e.
- We can see that Φ preserves the orbits.
  If Φ is a bijection, then the previous theorem is proved.

Injection

- Lemma : Φ is injective, and it is surjective if and only if λ satisfy the previous conditions.
- To see this, for any $\mathbf{w} \in \mathrm{rW}(\lambda)$, we check $\mathbf{w}^{(j)} = \mathbf{w}.\tau_{o(1)} \cdots \tau_{o(j)}$ for each j, and see how many i satisfy that $\mathbf{w}^{(i-1)}$ has a braid on position i.
  That is, $\mathbf{w}_{i-1}^{(i-2)}\mathbf{w}_{i}^{(i-2)}\mathbf{w}_{i+1}^{(i-2)} = a(a+1)a$ for some a.
- Take $\mathbf{w} = 1231423121 \in \mathrm{Red}(\mathbf{w}_0)$ as an example, we can write down the following chart

i	$\mathbf{w}^{(i-2)}$	a_i	c_i	$c_i - a_i$
2	1231423121	1	3	2
3	1213241321	2	3	1
4	1213214321	1	2	1
5	1231214321	1	1	0
6	1231241321	2	1	−1
7	1213423121	2	1	−1
8	1213423121	3	2	−1
9	1231241321	3	1	−2

where $a_i := \mathbf{w}_{i-1}^{(i-2)}$ and $c_i := \mathbf{w}_{i+1}^{(i-2)}$

Uniqueness of i

- We want to prove that at most one i satisfy $\mathbf{w}_{i-1}^{(i-2)} = \mathbf{w}_{i+1}^{(i-2)}$,, note that this is equivalent to a braid because if it is aba with $|a-b| \neq 1$, then we can swap a, b can cancel the double a to shorten the word, contradict that it is shortest.
- Lemma 5.4 : Let $\mathbf{w}' = \mathbf{w}.\tau_{o(i+1)}$, then

$$\mathbf{w}_{i-1} < \mathbf{w}_{i+1} \iff \mathbf{w}'_i \leq \mathbf{w}'_{i+2}$$

  Which can be proved by simply discussing whether $\mathbf{w}_{i-1}$ swap with $\mathbf{w}_i$ and whether $\mathbf{w}_{i+1}$ swap with $\mathbf{w}_{i+2}$.
- Proof of the uniqueness of i: if $\mathbf{w}_{i-1}^{(i-2)} = \mathbf{w}_{i+1}^{(i-2)}$, , and $\mathbf{w}^{(i-1)} = \mathbf{w}^{(i-2)}.\tau_{o(i-1)} = \mathbf{w}^{(i-2)}.\tau_{o(i+1)}$, so the previous lemma means that $\mathbf{w}_{i-1}^{(i-2)} = \mathbf{w}_{i+1}^{(i-2)} \implies \mathbf{w}_i^{(i-1)} > \mathbf{w}_{i+2}^{(i-1)}$ and similarly implies $\mathbf{w}_{i+1}^{(i)} = \mathbf{w}_{i+3}^{(i)}$ and so on, so no more j satisfies $\mathbf{w}_{j-1}^{(j-2)} = \mathbf{w}_{j+1}^{(j-2)}$.

Existence of i

- Now we prove that if $\lambda_1 > \lambda_2$ and $\lambda_\ell = 1$, then $\mathbf{w}_{i-1}^{(i-2)} = \mathbf{w}_{i+1}^{(i-2)}$ for some i.
- $\lambda_1 > \lambda_2 \iff \mathbf{w}_1 \leq \mathbf{w}_3$ for any $\mathbf{w} \in \mathrm{rW}(\lambda)$.
- $\lambda_\ell = 1 \iff \mathbf{w}_{|\lambda|-2} \geq \mathbf{w}_{|\lambda|}$ for any $\mathbf{w} \in \mathrm{rW}(\lambda)$.
- Hence for any word $\mathbf{w} \in \mathrm{rW}(\lambda)$, $\mathbf{w}_1^{(0)} \leq \mathbf{w}_3^{(0)}$ and $\mathbf{w}_{|\lambda|-2}^{(i-3)} \geq \mathbf{w}_{|\lambda|}^{(i-3)}$
- By the lemma on previous page, when we iterate the values of i from 2 to $|\lambda| - 1$, we can jump from $<$ to $>$ in one step, so the equality must hold at some point.
  Since that such i exists and is unique, Φ is bijective in this case.

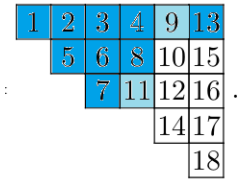
Homomesy for Posets

We provide a homomesy result of similar nature on poset.

- Let P be a finite poset with $|P| = n$, and let $\mathcal{L}(P)$ be the set of linear extensions of P.
- Let I be an order ideal of P, that is, for any $x \in I$ and $y \in P$, if $y < x$ then $y \in I$
- Define

$$\mathrm{des}_I(L) := \{p \in I | p \lessdot L^{-1}(L(p)+1) \in I\}$$

  Which is called the **descents** of L.
- For example, let P be a rSYT and its linear extension L is as follows, and the colored area is I. Then $\mathrm{des}_I(L)$ is the set of green cells.



Theorem 5.7

- We can also define an action τ_i on $\mathcal{L}(P)$ as interchanging i and $i+1$ if it is allowed, and similarly define τ_o and τ_e.
- Theorem 5.7: If P has a minimal element $\hat{0}$ and a maximal element $\hat{1}$, then for a fix ordered ideal $I \subset P$, the size of $\mathrm{des}_I(L)$ is homomesic with respect to $\langle \tau_o, \tau_e \rangle$ orbits with expected value $= 1$.
- Similarly we present a $\langle \tau_o, \tau_e \rangle$ orbits-preserving bijection between $\mathcal{L}(P)$ and $\{(p, L) | L \in \mathcal{L}(P), p \in \mathrm{des}_I(L)\}$

Bijection

- For any $L \in \mathcal{L}(P)$, we first consider a sequence of linear extensions $L_1, \cdots L_n$ such that $L_1 := L$ and $L_{i+1} := L_i.\tau_{o(i)}$
- Let $p_1, \cdots p_n \in P$ such that for each $i, p_i = L_i^{-1}(i)$, hence $p_1 = \hat{0}$ and $p_n = \hat{1}$.
- Consider L_i and L_{i+1}, if $\tau_{o(i)}$ swaps i and $i+1$, then $p_i = L_i^{-1}(i) = L_{i+1}^{-1}(i+1) = p_{i+1}$, if not, that means i is covered by $i+1$ in both L_i, and their position remain in L_{i+1}, therefore $p_i \lessdot p_{i+1}$.
- This fact implies that $p_1, \cdots p_n$ form a path from $\hat{0}$ to $\hat{1}$., and there must exist exactly one i such that $p_i \in I$ but $p_{i+1} \notin I$, and at this point $p_i \lessdot p_{i+1} = L_i^{-1}(L_i(p_i) + 1)$, so $p_i \in \mathrm{des}_I(L_i)$, and clearly no other j satisfy the condition.
- Define $\Phi : \{(p, L) | L \in \mathcal{L}(P), p \in \mathrm{des}_I(L)\} \to \mathcal{L}(P)$ such that

$$\Phi(p, L) = L.\tau_{o(L(p))}\tau_{o(L(p)-1)} \cdots \tau_{o(1)}$$

Then Φ is a $\langle \tau_o, \tau_e \rangle$ orbit preserving bijection.

Thank You for Listening